

1 Lecture 06 - Self-Evaluated Integration Case: Measuring Risk, Return, and Empirical Dependence

1.1 Purpose of the Case

No submission is expected for this case: it is self-reflection work designed to help you practice, check your reasoning, and identify what you may want to review.

This lecture is a self-evaluated integration case rather than a standard lecture chapter. Its purpose is to help you bring together the ideas from the first part of the course in one realistic data-analysis task.

Working through this case is beneficial because it forces several concepts to be used together:

- transforming prices into returns;
- summarizing a numerical distribution;
- interpreting volatility and downside risk;
- comparing empirical evidence with a probability benchmark;
- using conditional probabilities to think about dependence;
- writing conclusions in careful statistical language.

The case requires an external dataset and is best completed in a Python notebook. For that reason, the full lecture notes do not reproduce every calculation step here. Instead, this chapter states the case and explains what you should try to produce. After attempting the case, you can use the worked solution and example report in the solutions booklet for self-evaluation.

1.2 Case Scenario

You have a file containing daily adjusted closing prices for Meta Platforms from 2018 to July 2026:

`Lecture 06 - meta daily 2018 2026.csv`

The statistical question is not whether Meta Platforms is a good investment and not whether its future returns can be forecast from this sample. The goal is narrower and more useful for this course:

Use one historical dataset to write a disciplined statistical description of return, risk, distribution shape, tail behavior, and empirical dependence.

Use the `Adj Close` column to compute simple daily returns:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}.$$

1.3 What to Practice

When working on the case, try to answer the following questions in your notebook and in a short self-evaluation memo:

1. What is the date range of the return sample, and why is there one fewer return observation than price observation?
2. What are the mean, median, standard deviation, minimum, maximum, and key percentiles of daily returns?
3. What share of days had negative returns?
4. What does a histogram suggest about the return distribution?
5. How does the empirical lower tail compare with a normal benchmark using the sample mean and standard deviation?
6. Does the probability of a loss day change much after a previous loss day?
7. Does the probability of a large-loss day change after a previous large-loss day?

8. What conclusions can be stated carefully, and what conclusions would overinterpret the historical sample?

1.4 Expected Outcome

By the end of the case, you should be able to produce a short, self-evaluated risk-return note. It should explain the data, the return construction, the main descriptive statistics, the downside-risk evidence, the normal-benchmark comparison, and the empirical dependence check.

The important point is not to obtain a single "right" investment conclusion. The important point is to show that you can connect Python calculations with statistical interpretation.